DATA BINARIES

10 SQL patterns interviews keep asking

Tutorials teach one command at a time. Interviews mix them. These 10 patterns cover the questions that come up again and again for Data Analyst and Data Engineer roles. One pattern a day, ten days.

01 Join 2–3 tables

02 GROUP BY + HAVING

03 Rows with no match

04 Find duplicates

05 CASE WHEN

06 Filter by dates

07 Second highest value

08 Top N in each group

09 Running totals and rolling averages

10 Month-on-month growth

How to use this sheet

1. One pattern a day. Read the example, then close it and write the query yourself.
2. Give each question 20 minutes before you open a hint.
3. When it runs, say in one sentence what the result means for the business. Interviewers ask this next.
4. Can't solve the Level up question yet? Come back to it after Day 10.

DAY 01

Join 2–3 tables

Most interview answers need data from more than one table, like members, bookings and facilities. You must know which column links each pair of tables.

```
SELECT m.firstname, f.name
FROM members m
JOIN bookings b   ON b.memid = m.memid
JOIN facilities f ON f.facid = b.facid;
```

Interview tip: Before writing the query, say which column connects the tables. Interviewers listen for this.

PRACTISE

Members who have used a tennis court

PG Exercises ↗

LEVEL UP

Salaries Differences Easy

StrataScratch ↗

DAY 02

GROUP BY + HAVING

Totals per category, then keep only the groups that cross a limit: "facilities with more than 1,000 slots booked".

```
SELECT facid, SUM(slots) AS total_slots
FROM bookings
GROUP BY facid
HAVING SUM(slots) > 1000;
```

Interview tip: WHERE filters rows before grouping. HAVING filters groups after grouping. This difference is asked very often.

PRACTISE

Facilities with more than 1000 slots booked

PG Exercises ↗

LEVEL UP

Data Science Skills Easy

DataLemur · LinkedIn ↗

DAY 03

Rows with no match

"Customers who never ordered", "pages with no likes". An inner join hides these rows, so you need a LEFT JOIN.

```
SELECT p.page_id
FROM pages p
LEFT JOIN page_likes l ON l.page_id = p.page_id
WHERE l.page_id IS NULL;
```

Interview tip: LEFT JOIN keeps every row from the left table. Where nothing matched, the right side is NULL, and that is what you filter on.

PRACTISE

Page With No Likes **Easy**

DataLemur · Facebook ↗

LEVEL UP

Signup Activation Rate **Medium**

DataLemur · TikTok ↗

DAY 04

Find duplicates

Real data is messy. "Find companies that posted the same job twice" or "keep only the latest record" checks if you can clean data.

```
SELECT company_id, title, COUNT(*)
FROM job_listings
GROUP BY company_id, title
HAVING COUNT(*) > 1;
```

Interview tip: Group by every column that makes two rows "the same", then count. Ask the interviewer which columns define a duplicate.

PRACTISE

Duplicate Job Listings **Easy**

DataLemur · LinkedIn ↗

LEVEL UP

Finding Updated Records **Easy**

StrataScratch · Microsoft ↗

DAY 05

CASE WHEN

Turn values into categories or count only some rows: laptop views vs mobile views, calls that were international.

```
SELECT
  SUM(CASE WHEN device_type = 'laptop' THEN 1 ELSE 0 END) AS laptop_views,
  SUM(CASE WHEN device_type IN ('tablet','phone') THEN 1 ELSE 0 END) AS mobile_views
FROM viewership;
```

Interview tip: CASE inside SUM or COUNT lets you build several counts in one pass over the table.

PRACTISE

Laptop vs. Mobile Viewership Easy

DataLemur · NY Times ↗

LEVEL UP

International Call Percentage Medium

DataLemur · Verizon ↗

DAY 06

Filter by dates

"Users who sent the most messages in August", "gap between first and last post this year". Dates appear in almost every business question.

```
SELECT sender_id, COUNT(*) AS messages
FROM messages
WHERE sent_date >= '2022-08-01'
  AND sent_date <  '2022-09-01'
GROUP BY sender_id;
```

Interview tip: Use >= the start date and < the next period's start. It never misses rows from the last day, whatever the time.

PRACTISE

Teams Power Users Easy

DataLemur · Microsoft ↗

LEVEL UP

Average Post Hiatus (Part 1) Easy

DataLemur · Facebook ↗

DAY 07

Second highest value

The classic: second highest salary, third transaction of each user. It checks subqueries and ranking.

```
SELECT MAX(salary) AS second_highest
FROM employee
WHERE salary < (SELECT MAX(salary) FROM employee);
```

Interview tip: Know two ways: the subquery above, and ranking with DENSE_RANK(). Say why DENSE_RANK handles equal salaries correctly.

PRACTISE

Second Highest Salary **Medium**

DataLemur · FAANG ↗

LEVEL UP

User's Third Transaction **Medium**

DataLemur · Uber ↗

DAY 08

Top N in each group

"Top 3 salaries in each department", "two best-selling products in each category". One of the most asked window-function questions.

```
SELECT * FROM (
  SELECT name, department_id, salary,
    DENSE_RANK() OVER (PARTITION BY department_id
                       ORDER BY salary DESC) AS rnk
  FROM employee
) t
WHERE rnk <= 3;
```

Interview tip: PARTITION BY restarts the ranking for every group. You can't filter a window function in WHERE directly, so wrap it in a subquery.

PRACTISE

Top Three Salaries **Medium**

DataLemur · FAANG ↗

LEVEL UP

Highest-Grossing Items **Medium**

DataLemur · Amazon ↗

DAY 09

Running totals and rolling averages

"3-day rolling average of tweets", "revenue to date". Used in every dashboard, so analysts get asked this a lot.

```
SELECT user_id, tweet_date,
  ROUND(AVG(tweet_count) OVER (
    PARTITION BY user_id ORDER BY tweet_date
    ROWS BETWEEN 2 PRECEDING AND CURRENT ROW), 2) AS rolling_avg_3d
FROM tweets;
```

Interview tip: ROWS BETWEEN decides how many rows go into each average. For a running total, use SUM(...) OVER (ORDER BY date).

PRACTISE

Tweets' Rolling Averages **Medium**

DataLemur · Twitter ↗

LEVEL UP

Rolling average of total revenue

PG Exercises ↗

DAY 10

Month-on-month growth

"How much did revenue grow compared to last month?" This combines GROUP BY, dates and LAG, so it is often the hardest question in the round.

```
WITH monthly AS (
  SELECT DATE_TRUNC('month', created_at) AS month,
         SUM(value) AS revenue
  FROM transactions GROUP BY 1
)
SELECT month, revenue,
  ROUND((revenue - LAG(revenue) OVER (ORDER BY month)) * 100.0
        / LAG(revenue) OVER (ORDER BY month), 2) AS pct_change
FROM monthly;
```

Interview tip: LAG() reads the previous row. Build monthly totals first, then compare. Multiply by 100.0, not 100, so the division isn't rounded to a whole number.

PRACTISE

Monthly Percentage Difference **Hard**

StrataScratch · Amazon ↗

LEVEL UP

Y-on-Y Growth Rate **Hard**

DataLemur · Wayfair ↗

DATA BINARIES

Finished all 10?
Now practise speaking it.

Writing the query is half the interview. The other half is explaining your approach out loud while you write it. Pick any 3 questions from this sheet and solve them again while talking through each step, the way you would in front of an interviewer.

Need help with SQL or your Data Analyst / Data Engineer preparation?
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